Does Decision Frequency Matter? Evaluating LLM Trading Agents Across Daily, Weekly, and Monthly Horizons

Erik Ely and Idea-Explorer

Abstract

Large language model (LLM) trading agents have attracted significant research attention, yet virtually all published systems operate at daily decision frequency—and recent rigorous evaluations show they consistently underperform passive benchmarks over longer horizons. We ask a fundamental question: is the problem the LLM approach itself, or the daily frequency? We present the first systematic comparison of LLM trading agent performance across three decision frequencies—daily, weekly, and monthly—using GPT-4.1-mini on five stocks over 2024. Our key finding is that monthly LLM trading achieves comparable risk-adjusted returns to daily trading (mean Sharpe ratio 1.10 vs. 1.17) while reducing maximum drawdowns by 36% (8.9% vs. 14.0%) and API costs by 95%. Surprisingly, weekly trading performs worst (mean Sharpe 0.19), suggesting an intermediate-frequency "valley" where LLMs lose both daily momentum signals and monthly strategic context. At the monthly horizon, the LLM exhibits qualitatively different behavior: it adopts conservative, trend-following strategies that capture major moves while avoiding whipsaws. On two of five stocks, monthly LLM trading outperforms Buy-and-Hold (JPM: +57.0% vs. +42.8%; TSLA: +104.5% vs. +52.7%). These results suggest that the widely reported failure of daily LLM trading may partly reflect a frequency mismatch rather than a fundamental capability limitation, and that less frequent decision-making deserves greater attention from the research community.

1 Introduction

LLM-based trading agents represent one of the most actively studied applications of large language models, with over 70 papers published since 2023 [Ding et al., 2024]. These systems use LLMs to process financial news, price data, and technical indicators to generate autonomous trading decisions. Yet despite this rapid growth, a striking pattern has emerged: when subjected to rigorous, long-term evaluation, daily LLM trading agents *consistently underperform* simple passive benchmarks such as Buy-and-Hold [Li et al., 2026, DeepFund Team, 2025].

This underperformance has been attributed to several factors, including regime miscalibration [Li et al., 2026], data leakage in backtesting [DeepFund Team, 2025], and overfitting to short evaluation windows [Ding et al., 2024]. However, one variable has been entirely overlooked: **decision frequency**. Every published LLM trading system we surveyed—across 23 papers in our literature review—operates exclusively at daily frequency. No prior work has systematically tested whether LLMs might perform better when making less frequent, more strategic decisions.

Why might frequency matter? LLMs excel at synthesizing information, identifying patterns, and reasoning about complex scenarios. These capabilities align naturally with strategic, longer-horizon decisions where fundamental trends dominate over noise. Daily trading, by contrast, demands rapid pattern recognition in noisy data—a task where traditional quantitative methods have well-established advantages. If LLMs are better suited to strategic reasoning than tactical execution, the universal default to daily trading may represent a fundamental frequency mismatch.

Evidence supports this intuition. DeepFund Team [2025] found that the only profitable LLM strategy in live trading used conservative, low-frequency positioning. Li et al. [2026] diagnosed *regime miscalibration* as a key failure mode—LLMs are overly conservative in bull markets and overly aggressive in bear markets—suggesting that daily noise may overwhelm their ability to identify market regimes. Li et al. [2024] found that ETF trading (the most strategic, long-term task) was the hardest for LLMs yet improved most with capable models, hinting that longer-horizon reasoning requires and rewards stronger capabilities.

We present the first systematic comparison of LLM trading performance across three decision frequencies: daily, weekly, and monthly. Using GPT-4.1-mini on five stocks (AAPL, JPM, NVDA, TSLA, MSFT) during 2024, we control for stock selection, time period, model, and information structure, isolating decision frequency as the sole independent variable. Our results reveal that:

- Monthly LLM trading achieves a mean Sharpe ratio of 1.10, comparable to daily trading (1.17), while reducing average maximum drawdowns by 36% (8.9% vs. 14.0%) and API costs by 95%.
- Weekly trading performs worst (mean Sharpe 0.19), revealing an intermediate-frequency "valley" where LLMs lack both daily momentum signals and monthly strategic context.
- At monthly frequency, the LLM exhibits qualitatively different behavior—adopting trend-following strategies with lower turnover—and outperforms Buy-and-Hold on 2 of 5 stocks, a rare result in the literature.

These findings suggest that the widely reported underperformance of LLM trading agents may partly be a frequency problem, not a capability problem. We discuss limitations of our study—including a small sample of 5 stocks and a single evaluation year—and outline directions for larger-scale validation.

The remainder of this paper is organized as follows. Section 2 reviews related work on LLM trading agents. Section 3 describes our experimental methodology. Section 4 presents results and analysis. Section 5 discusses implications and limitations, and section 6 concludes.

2 Related Work

LLM trading agent architectures. The design of LLM trading agents has progressed rapidly from simple prompt-based systems to sophisticated multi-agent architectures. FINMEM [Yu et al., 2023] introduced a layered memory architecture with shallow (daily), intermediate (quarterly), and deep (annual) memory, achieving strong short-term results on individual stocks. FINCON [Yu et al., 2024] extended this with a manager-analyst hierarchy and Conceptual Verbal Reinforcement, showing particular strength in bearish markets. TRADINGAGENTS [Xiao et al., 2025] mirrors the organizational structure of real trading firms with specialized analyst, researcher, and risk management agents. Despite these architectural advances, all three systems—and every other published LLM trading agent we surveyed—operate exclusively at daily decision frequency.

Rigorous evaluation of LLM trading. Recent work has challenged the optimistic results reported in earlier studies. FINSABER [Li et al., 2026] introduced a bias-mitigated backtesting framework that evaluates LLM strategies over 20 years across 100+ stocks, finding that strategies appearing profitable in short-term evaluations (6 months, 4–8 stocks) consistently underperform Buy-and-Hold at scale. The key diagnosed failure mode is *regime miscalibration*: LLMs are overly conservative in bull markets and overly aggressive in bear markets. DEEPFUND [DeepFund Team, 2025] complemented this with live (non-backtested) evaluation, finding that only 1 of 9 LLMs achieved positive returns over 24 trading days—and that model succeeded via conservative cash management rather than superior prediction. STOCKBENCH [Chen et al., 2025] constructed a contamination-free benchmark on post-training-cutoff data, finding that most LLMs only marginally beat Buy-and-Hold while providing better risk management. These findings collectively suggest that daily LLM trading faces fundamental challenges, but none of these studies considered alternative decision frequencies.

Reinforcement learning approaches. An alternative line of work uses reinforcement learning to fine-tune LLMs for trading. FLAG-TRADER [Li et al., 2025] showed that a 135M-parameter LLM fine-tuned with PPO outperforms GPT-4 in agentic frameworks, using Sharpe ratio change as the RL reward. While RL-based approaches address some limitations of pure prompting, they still operate at daily frequency and require substantial computational resources for training. Our work is complementary: we investigate whether the *decision horizon* itself is a key variable, independent of the agent architecture.

Multi-frequency analysis in traditional finance. The idea that trading performance depends on decision frequency is well-established in traditional finance. Momentum strategies show frequency-dependent returns [Jegadeesh and Titman, 1993], and optimal rebalancing frequency varies by asset class and market regime. However, this dimension has not been explored for LLM-based agents. Our work bridges this gap by applying the lens of frequency analysis to the emerging field of LLM trading.

Benchmarks and surveys. INVESTORBENCH [Li et al., 2024] provides a benchmark testing 13 LLMs across stocks, crypto, and ETFs, finding that ETF trading—the most strategic, long-term task—was hardest for most LLMs but improved most with capable models. Ding et al. [2024] surveyed 14 LLM trading systems and found a median backtesting period of only 1.3 years, with trading costs almost universally ignored. Both observations motivate our investigation of longer-horizon strategies that naturally reduce trading frequency and costs.

3 Methodology

We design a controlled experiment that isolates decision frequency as the sole independent variable while holding constant the LLM backbone, stock universe, evaluation period, and information structure.

3.1 Problem Formulation

At each decision point t, the LLM agent receives a context window $\boldsymbol{c}_t$ containing recent price history, technical indicators, and the current portfolio position. It outputs a discrete action $a_t \in \{\text{BUY}, \text{HOLD}, \text{SELL}\}$. The portfolio follows binary positioning: 100% long or 100% cash (no shorting, no partial positions). A BUY action enters a full long position, SELL exits to cash, and HOLD maintains the current state. The starting position is cash for all experiments.

We evaluate three decision frequencies $f \in \{\text{daily}, \text{weekly}, \text{monthly}\}$, where the number of decision points per stock varies accordingly: 252 (daily), 53 (weekly), and 12 (monthly) during the evaluation year.

3.2 Data

Source and coverage. We use daily, weekly, and monthly OHLCV (Open, High, Low, Close, Volume) data from Yahoo Finance via the `yfinance` library, spanning February 2016 to December 2024. The first eight years (2016–2023) provide historical context for technical indicator computation, while 2024 serves as the evaluation period.

Stock universe. We select five stocks spanning diverse sectors and volatility profiles: AAPL (technology), JPM (finance), NVDA (semiconductors), TSLA (automotive/energy), and MSFT (technology). This selection balances sector diversity with coverage of both high-volatility (TSLA, NVDA) and lower-volatility (AAPL, JPM, MSFT) equities. During 2024, SPY returned approximately 25%, making this a generally bullish evaluation period with sector-specific pullbacks.

Technical indicators. At each frequency, we compute the following indicators over the corresponding period granularity:

- **SMA-10 and SMA-20**: 10-period and 20-period simple moving averages, providing trend signals.
- **RSI-14**: 14-period relative strength index, indicating overbought/oversold conditions.
- **Returns**: 1-period and 5-period percentage returns.

Information window. At each decision point, the LLM receives the most recent 15 periods of data at the corresponding frequency. This means daily agents see 15 trading days ($\sim$3 weeks), weekly agents see 15 weeks ($\sim$3.5 months), and monthly agents see 15 months ($\sim$1.25 years). The information window naturally scales with frequency, mirroring how human investors adjust their analysis horizon to their trading horizon.

3.3 LLM Agent Design

Model. We use OpenAI's GPT-4.1-mini as the LLM backbone for all experiments. This model balances capability with cost-effectiveness, enabling the 1,585 total API calls required across all conditions within a budget of $1.34.

Prompt design. The agent receives a standardized system prompt instructing it to act as a stock trading analyst and output a JSON object with two fields: `decision` (BUY, HOLD, or SELL) and `reasoning` (a brief explanation). The user prompt contains:

1. The stock ticker and decision frequency context (e.g., "You are making a *monthly* trading decision").
2. The current position (LONG or OUT).
3. A frequency-specific instruction (e.g., "Your next decision will be in one month").
4. A table of recent price history with Close, Volume, SMA-10, SMA-20, RSI-14, 1-period return, and 5-period return.

The prompt is identical across frequencies except for the frequency context, the instruction about time until next decision, and the data granularity. This ensures that any performance differences are attributable to frequency rather than prompt variation.

Inference configuration. We use temperature 0.0 for deterministic outputs, with a maximum of 150 output tokens. Each experiment constitutes a single run per condition, producing one trajectory per stock-frequency pair.

3.4 Baselines

Buy-and-Hold (B&H). The passive benchmark that buys at the start of the evaluation period and holds throughout. This is the most commonly used baseline in LLM trading research [Ding et al., 2024].

SMA Crossover (SMA-CROSSOVER). A simple technical strategy that buys when the 10-period SMA crosses above the 20-period SMA and sells on the reverse crossover. This provides a non-LLM active trading baseline.

3.5 Evaluation Metrics

We evaluate performance using three standard metrics:

- **Cumulative Return (CR)**: Total percentage gain or loss over the evaluation period.
- **Sharpe Ratio (SR)**: Annualized risk-adjusted return, computed as the mean return divided by its standard deviation, scaled by $\sqrt{N}$ where N is the annualization factor (252 for daily, 52 for weekly, 12 for monthly).
- **Maximum Drawdown (MDD)**: The largest peak-to-trough decline, measuring worst-case downside risk.

We additionally report decision distributions (percentage of BUY, HOLD, and SELL actions) and turnover rates (percentage of periods with position changes) to characterize agent behavior.

3.6 Statistical Analysis

We use paired t-tests to compare Sharpe ratios across frequencies, with each stock providing one paired observation. We apply Bonferroni correction for three pairwise comparisons (daily vs. weekly, daily vs. monthly, weekly vs. monthly) at $\alpha = 0.05$. We report Cohen's d as a measure of effect size. With only $n = 5$ paired observations, we acknowledge limited statistical power and emphasize effect sizes alongside p-values.

Table 1: Sharpe ratios by stock and decision frequency. Best LLM frequency per stock in **bold**. Monthly trading achieves comparable mean performance to daily while weekly trading lags substantially.

Stock	DAILY	WEEKLY	MONTHLY	B&H
AAPL	**1.38**	0.11	1.24	1.20
JPM	0.60	0.31	**2.39**	1.61
NVDA	**2.28**	1.26	0.36	3.47
TSLA	0.86	0.75	**2.10**	0.75
MSFT	**0.70**	−1.50	−0.60	0.44
Mean	1.17	0.19	1.10	1.49

Table 2: Cumulative returns (%) by stock and decision frequency. Best LLM frequency per stock in **bold**. Monthly trading achieves the highest mean cumulative return among LLM frequencies and outperforms B&H on JPM and TSLA.

Stock	DAILY	WEEKLY	MONTHLY	B&H
AAPL	**+24.4**	+7.1	+23.2	+32.0
JPM	+12.8	+10.1	**+57.0**	+42.8
NVDA	**+81.5**	+46.1	+16.3	+187.2
TSLA	+37.9	+45.6	**+104.5**	+52.7
MSFT	**+14.5**	−17.4	−2.7	+13.7
Mean	+34.2	+18.3	+39.7	+65.7

4 Results

4.1 Main Results: Risk-Adjusted Performance

table 1 presents Sharpe ratios across all stock-frequency combinations. The central finding is that monthly trading achieves a mean Sharpe ratio (1.10) comparable to daily trading (1.17), while weekly trading performs substantially worse (0.19). None of the LLM frequencies match the Buy-and-Hold mean Sharpe of 1.49 on average, though monthly LLM trading outperforms B&H on two individual stocks.

The stock-level results reveal an important pattern: monthly trading excels on stocks with strong directional trends (JPM, TSLA) while daily trading performs better on stocks requiring finer-grained timing (AAPL, NVDA, MSFT). On JPM, the monthly agent achieves a Sharpe ratio of 2.39 versus 0.60 for daily—a nearly 4× improvement. On TSLA, monthly achieves 2.10 versus 0.86 for daily. These are precisely the stocks where the monthly agent adopted a “buy and hold on strength” strategy, as we analyze in section 4.5.

4.2 Cumulative Returns

table 2 presents cumulative returns. Monthly trading achieves the highest mean return (39.7%) among the three LLM frequencies, surpassing daily (34.2%) and substantially outperforming weekly (18.3%). However, all LLM frequencies underperform B&H (65.7%) on average, consistent with prior findings in the literature [Li et al., 2026].

The most striking result is TSLA at monthly frequency: the LLM achieves +104.5% cumulative return, nearly doubling the Buy-and-Hold return of +52.7% and nearly tripling the daily LLM return of +37.9%. Similarly, monthly JPM returns +57.0% compared to Buy-and-Hold’s +42.8%. These results demonstrate that LLM agents *can* outperform passive investing when operating at an appropriate frequency, even in the absence of news or sentiment data.

Table 3: Maximum drawdown (%) by stock and decision frequency. Best (smallest magnitude) per stock in **bold**. Monthly trading achieves the lowest average drawdown, reducing it by 36% relative to daily and 59% relative to B&H.

Stock	DAILY	WEEKLY	MONTHLY	B&H
AAPL	−9.0	−12.9	−**5.8**	−15.4
JPM	−7.9	−9.4	−**6.2**	−10.1
NVDA	−20.5	−21.8	−**13.1**	−27.0
TSLA	−24.2	−23.6	−**7.7**	−40.9
MSFT	−**8.6**	−18.3	−11.9	−15.5
Mean	−14.0	−17.2	−**8.9**	−21.8

Table 4: Statistical comparison of Sharpe ratios across frequencies. We report paired t-test statistics, raw and Bonferroni-corrected p-values, and Cohen's d. The large effect sizes for daily vs. weekly ($d = 1.31$) and weekly vs. monthly ($d = 0.92$) suggest practically meaningful differences despite limited sample size.

Comparison	t-statistic	p-value	p (Bonferroni)	Cohen's d
DAILY VS. WEEKLY	2.61	0.059	0.178	1.31
DAILY VS. MONTHLY	0.09	0.931	1.000	0.05
WEEKLY VS. MONTHLY	−1.85	0.138	0.415	0.92

4.3 Risk Management: Maximum Drawdown

table 3 reveals the most consistent advantage of monthly trading: superior drawdown control. Monthly decisions achieve the lowest average maximum drawdown (−8.9%) compared to daily (−14.0%), weekly (−17.2%), and Buy-and-Hold (−21.8%). This advantage holds across 4 of 5 stocks.

The TSLA result is particularly dramatic: monthly MDD is only −7.7%, compared to −24.2% for daily and −40.9% for Buy-and-Hold. This represents a 5.3× drawdown reduction relative to Buy-and-Hold, suggesting that the monthly LLM agent effectively avoided TSLA's major pullbacks during 2024.

4.4 Statistical Significance

table 4 presents paired t-test results comparing Sharpe ratios across frequencies. No comparison reaches statistical significance after Bonferroni correction, which is unsurprising given only $n = 5$ paired observations. However, the effect sizes are informative.

The daily-vs.-weekly comparison yields a large effect size (Cohen's $d = 1.31$) with an unadjusted p-value of 0.059, approaching conventional significance. The weekly-vs.-monthly comparison also shows a large effect ($d = 0.92$). In contrast, daily-vs.-monthly shows negligible effect ($d = 0.05$), confirming that these two frequencies produce similar risk-adjusted returns. A power analysis suggests that 15–20 stocks would be sufficient to detect the daily-vs.-weekly effect at $\alpha = 0.05$ with 80% power.

4.5 Behavioral Analysis

table 5 presents the decision distributions and turnover rates across frequencies, revealing qualitatively different behavioral patterns.

Daily decisions: noise reactivity. At daily frequency, the agent shows high turnover (42%) with frequent oscillation between BUY and SELL. Reasoning outputs cite short-term indicators such as "recent pullback" and "one-day decline," suggesting the model struggles to distinguish signal from noise at this granularity.

Table 5: Decision distributions and turnover rates across frequencies, averaged over all five stocks. Monthly trading shows the highest HOLD rate and lowest turnover, indicating more conservative, trend-following behavior.

Frequency	BUY (%)	HOLD (%)	SELL (%)	Turnover (%)
DAILY	15	49	36	42
WEEKLY	25	43	31	53
MONTHLY	25	50	25	40

Table 6: API cost comparison across frequencies. Token counts and costs reflect all five stocks over the 2024 evaluation period using GPT-4.1-mini.

Frequency	API Calls	Total Tokens	Cost (USD)
DAILY	1,260	1,403,913	$1.06
WEEKLY	265	297,871	$0.23
MONTHLY	60	67,130	$0.05

Weekly decisions: worst of both worlds. Weekly trading exhibits the *highest* turnover (53%) with the poorest performance. The agent appears caught between time horizons—too slow to capture daily momentum, too fast for trend following. This produces a "worst of both worlds" effect where the model reacts to weekly noise without the strategic context that monthly data provides.

Monthly decisions: strategic positioning. At monthly frequency, the LLM adopts distinctly different behavior. The most striking example is JPM, where the agent issued 0 SELL decisions (83% HOLD), effectively buying in January 2024 and riding the year-long uptrend. On TSLA, the agent strategically avoided the Q1–Q2 downturn via HOLD/SELL decisions and entered before the Q4 rally with BUY signals in June and November. The reasoning outputs cite "long-term uptrend," "macro patterns," and "bullish crossover"—qualitatively different from the short-term language used at daily frequency.

4.6 API Cost Analysis

Decision frequency has a direct and dramatic impact on API costs. table 6 shows that daily trading consumed 1,403,913 tokens ($1.06) across all five stocks, while monthly trading used only 67,130 tokens ($0.05)—a 21× cost reduction. For multi-agent systems like TRADINGAGENTS, which require 11+ LLM calls per decision, this cost differential would be even more pronounced.

5 Discussion

5.1 Interpretation of Results

The frequency–performance relationship is non-monotonic. Our results challenge the intuitive expectation that performance improves monotonically as decision frequency decreases. Instead, we observe a U-shaped pattern: daily and monthly trading perform comparably (mean Sharpe 1.17 and 1.10), while weekly trading performs substantially worse (0.19). This suggests that intermediate frequencies may represent a "valley of indecision" where LLMs lose the benefits of both high-frequency momentum signals and low-frequency strategic context.

Monthly trading enables qualitatively different strategies. The behavioral analysis reveals that the monthly LLM does not simply make fewer of the same decisions—it makes *different kinds* of decisions. At monthly frequency, the agent demonstrates strategic patience (83% HOLD on JPM), selective market timing (avoiding TSLA's Q1–Q2 drawdown), and trend-following logic in its reasoning. This qualitative shift suggests that longer decision horizons allow LLMs to leverage their strength in strategic reasoning rather than forcing them into a tactical role where they are disadvantaged.

Risk management is the most consistent advantage. While the Sharpe ratio comparison between daily and monthly is approximately equal, the drawdown advantage of monthly trading is consistent across 4 of 5 stocks. The mean MDD improvement from −14.0% (daily) to −8.9% (monthly) represents a 36% reduction, and the improvement relative to Buy-and-Hold (−21.8%) is 59%. For practitioners concerned with downside risk, this consistent advantage may be more important than marginal Sharpe ratio differences.

5.2 Why Does Weekly Trading Fail?

The poor performance of weekly trading is one of our most surprising findings. We hypothesize three potential explanations:

First, weekly data may occupy an informationally "thin" regime. Daily data captures short-term momentum and mean-reversion patterns; monthly data reveals macro trends and regime changes. Weekly data may be too noisy for trend identification yet too coarse for momentum capture.

Second, the weekly agent exhibits the highest turnover rate (53%), suggesting it overreacts to weekly fluctuations. Unlike daily noise—which may average out over many rapid decisions—weekly overreaction leads to poorly timed trades with fewer opportunities for correction.

Third, prompt design may inadvertently disadvantage weekly decisions. The instruction "Your next decision will be in one week" may create urgency without providing sufficient context for confident positioning. Future work should investigate whether prompt engineering or richer information (such as news) can improve weekly performance.

5.3 Comparison to Prior Work

Our finding that LLM trading underperforms Buy-and-Hold on average (best LLM mean: 39.7% vs. B&H 65.7%) is consistent with FINSABER [Li et al., 2026] and DEEPFUND [DeepFund Team, 2025]. However, our result that monthly LLM trading *outperforms* B&H on 2 of 5 stocks (JPM: +57.0% vs. +42.8%; TSLA: +104.5% vs. +52.7%) is rare in the literature. This suggests that frequency optimization may be a viable path to competitive LLM trading strategies, particularly for stocks with strong directional trends or high volatility.

The "buy and hold on strength" behavior exhibited by the monthly JPM agent echoes the finding from DEEPFUND [DeepFund Team, 2025] that conservative strategies with high cash reserves were the only profitable approach in live trading. Both results point to the same conclusion: LLMs may add the most value when they trade infrequently and focus on high-conviction decisions.

5.4 Limitations

We identify seven limitations that should guide interpretation of our results and design of follow-up studies:

Small sample size. With only 5 stocks, we lack statistical power to detect moderate effect sizes. The large Cohen's d values (1.31 for daily vs. weekly) suggest real effects, but validation on 15–20 stocks is needed for robust conclusions.

Single evaluation year. Our results are limited to 2024, a generally bullish year. The hypothesis that monthly trading excels at regime detection is most interesting in bear or mixed markets, which we do not evaluate. Extended evaluation across multiple market regimes is a critical next step.

Price-only information. We deliberately exclude news and sentiment data to isolate the frequency effect. In practice, LLM trading agents typically process textual information, which could interact differently with decision frequency. News-augmented experiments would test whether the frequency effect persists with richer information.

Single model. We use only GPT-4.1-mini. More capable models (e.g., GPT-4.1, Claude 4.5 Sonnet) might show different frequency effects, particularly if their stronger reasoning capabilities benefit more from longer-horizon strategic decisions.

No transaction costs. We do not model transaction costs, which would further favor monthly trading due to its lower turnover. Including realistic transaction costs would strengthen the case for less frequent trading.

Deterministic inference. Using temperature 0.0 produces a single trajectory per condition, preventing estimation of variance. Stochastic evaluation with temperature 0.3 and multiple runs per condition would provide confidence intervals and strengthen statistical analysis.

Limited monthly data points. With only 12 monthly decision points per stock, the statistical reliability of monthly performance metrics is inherently limited. This is a structural challenge for any study of low-frequency trading strategies.

6 Conclusion

We present the first systematic study of decision frequency effects on LLM trading agent performance. By comparing daily, weekly, and monthly trading with GPT-4.1-mini across five stocks during 2024, we find that monthly decisions achieve comparable risk-adjusted returns to daily trading (mean Sharpe 1.10 vs. 1.17) while reducing maximum drawdowns by 36% and API costs by 95%. Weekly trading performs worst, revealing a non-monotonic frequency–performance relationship. At monthly frequency, the LLM exhibits qualitatively different strategic behavior—adopting trend-following approaches with lower turnover—and outperforms Buy-and-Hold on 2 of 5 stocks.

These results suggest that the widely reported underperformance of daily LLM trading may partly be a frequency mismatch rather than a fundamental capability limitation. The research community's universal default to daily decision-making deserves reconsideration, and future work on LLM trading agents should explore frequency as a first-class design variable alongside model architecture and information sources.

Future work. Three directions are most pressing: (1) scaling to 15–20 stocks across multiple years, including bear markets, to achieve statistical significance and test regime-specific hypotheses; (2) adding a biweekly frequency to map the "valley" between weekly and monthly performance; and (3) testing whether more capable LLMs and richer information sources (e.g., news, earnings) amplify or diminish the frequency effect.

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